

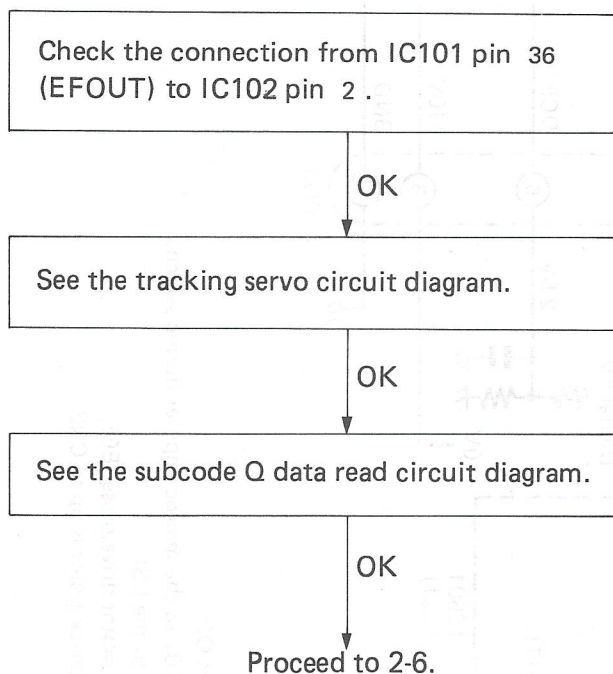
Spindle motor  
circuit board

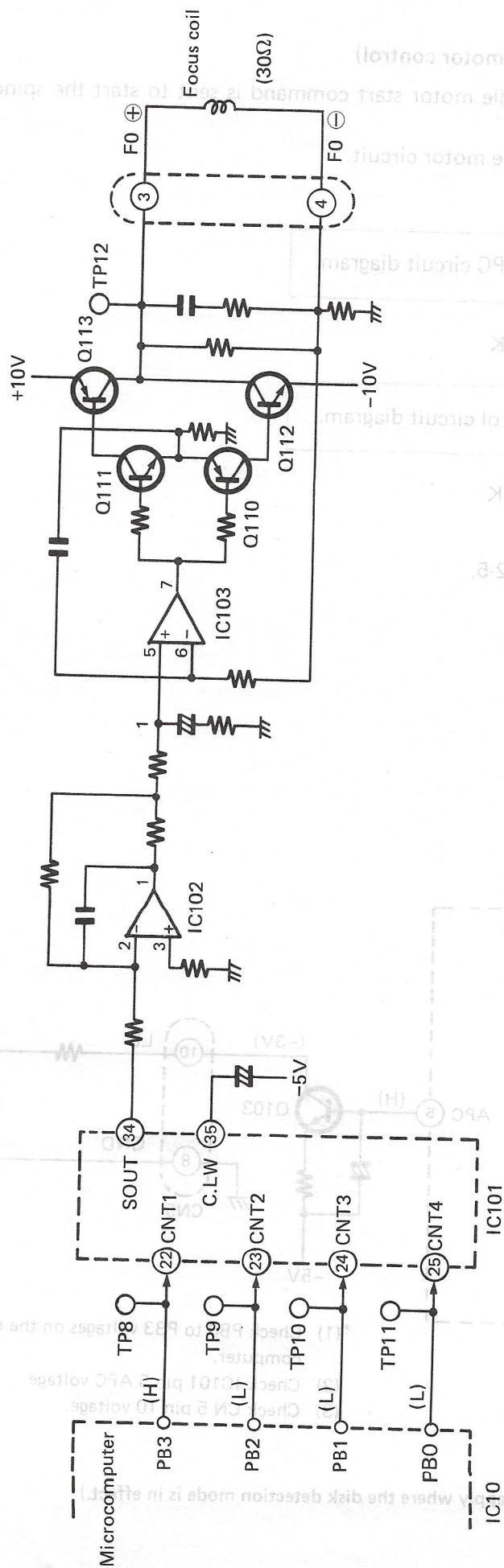
**Fig. 2-4-2 Spindle Motor Control Circuit (Circuit voltages apply where the play mode is in effect.)**

## 2-5 RF Signal Check (RF Signal, Tracking Servo, LSI)

- 1) When an RF signal is detected, the focus circuit changes from the search mode to the focus servo mode.
- 2) In this case, a focus lock signal is applied to the microcomputer, which sends a tracking ON, command to start the tracking servo circuit.
- 3) Then, the RF signal is converted to an EFM signal by the PLL IC, and the converted EFM signal is applied to the LSI. The LSI demodulates the subcode Q data and supplies it to the microcomputer, which deciphers the TOC and selects the play mode.

—If a fault is detected anywhere up to this point, the eject mode follows.—





Focus search takes place for disk detection. CNT1 is high due to a command (disk detection) from the microcomputer, IC101 34 outputs a search signal (triangular wave) to drive the focus coil.

Fig. 2-3-2 Focus Search Circuit (Circuit voltages apply where focus search is under way.)

## 2-4 Spindle Motor Circuit Check (Laser ON, spindle motor control)

When the focus is adjusted by focus search, a spindle motor start command is sent to start the spindle motor.

First, check the laser that it is ON and then the spindle motor circuit.

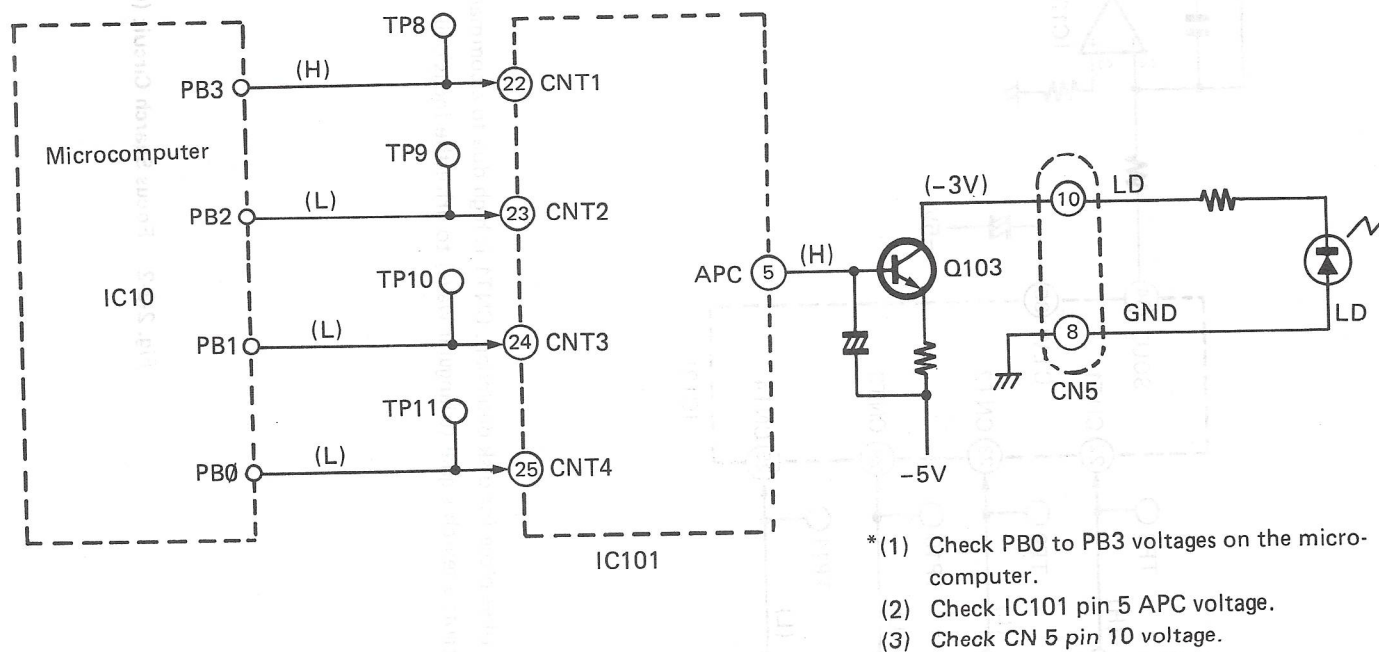
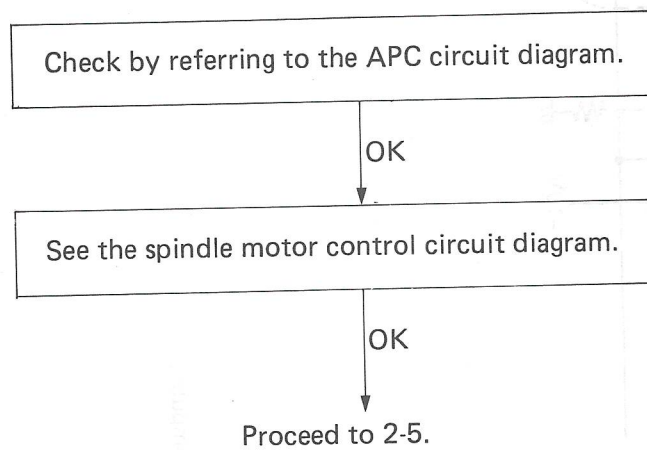


Fig. 2-4-1 APC Circuit (Circuit voltages apply where the disk detection mode is in effect.)

## 2-3 Focus Search Check (Traverse Servo, Focus Servo)

When a disk is set on the turntable, the traverse motor runs forward, and the focus starts searching. Proceed as follows:

- (1) Traverse motor forward run check
- (2) Focus search check

Fig. 3-3-1 Traverse Motor Forced Operation Circuit (Check whether the traverse motor is running forward)

3. If B83 shows no change, check the circuit from B10 and B11 to the traverse motor.

(Normally, observation starts in the internal switch ON state. The switch turns off when the traverse motor starts running forward.)

3. Check if microcomputer B83 changes from low to high.

1. Check microcomputer B83 (CMT1) test it is right.

and the forced mode ends when the observed pickup reaches the prescribed position.

Internal switch state detection by B83 on the microcomputer controls the traverse motor to run forward or backward.

forced traverse motor operation.

When a disk is set on the turntable, the microcomputer is set in the disk detection mode (CH1 high), and IC903 runs on.

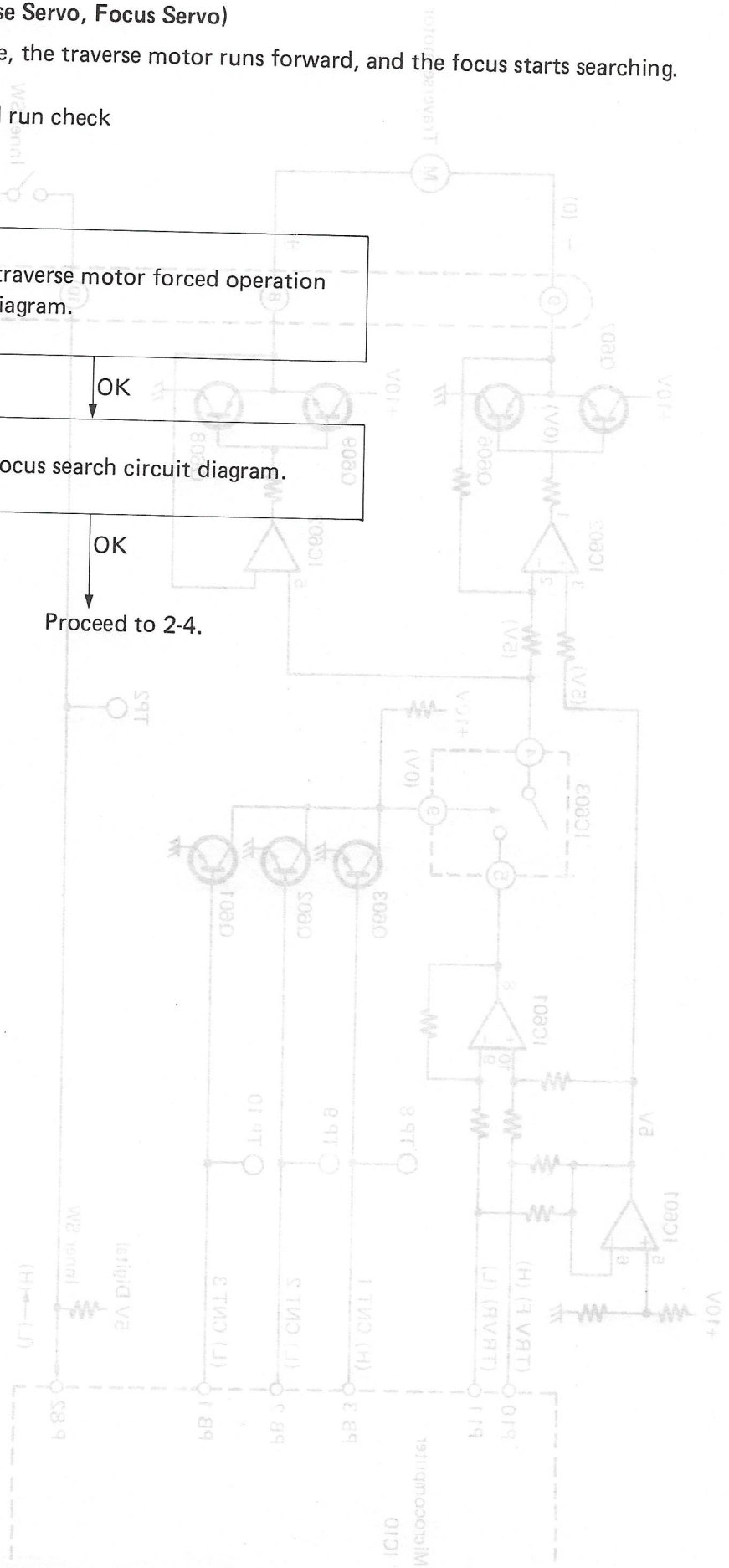
See the traverse motor forced operation circuit diagram.

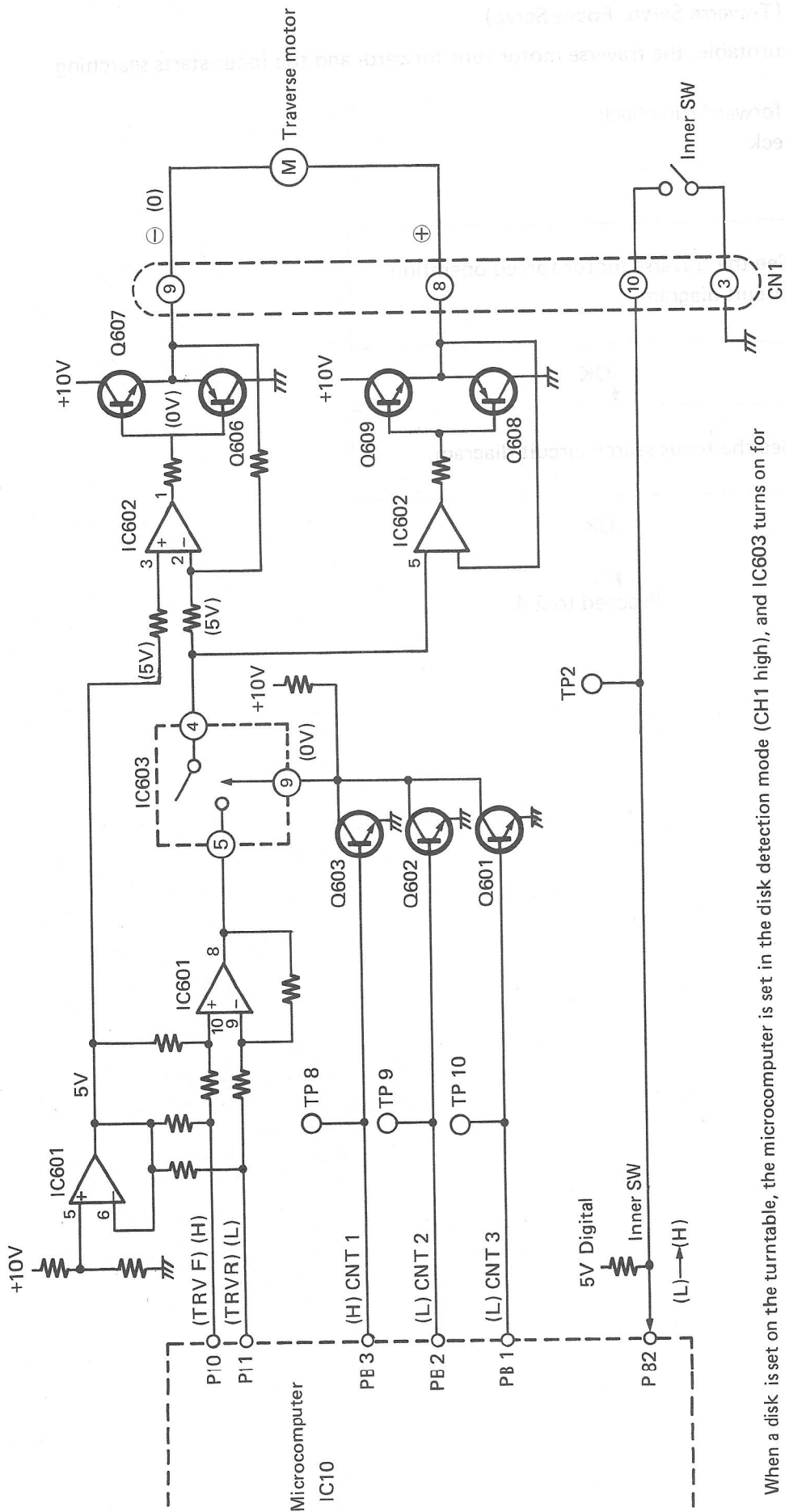
OK

See the focus search circuit diagram.

OK

Proceed to 2-4.





When a disk is set on the turntable, the microcomputer is set in the disk detection mode (CH1 high), and IC603 turns on for forced traverse motor operation.  
Internal switch state detection by P82 on the microcomputer controls the traverse motor to run forward or backward (P10, P11), and the forced mode ends when the optical pickup reaches the prescribed position.

1. Check microcomputer PB3 (CNT1) that it is high.
2. Check if microcomputer P82 changes from low to high.  
(Normally, operation starts in the internal switch ON state. The switch turns off when the traverse motor starts running forward.)
3. If P82 shows no change, check the circuit from P10 and P11 to the traverse motor.

Fig. 2-3-1 Traverse Motor Forced Operation Circuit (Circuit voltages apply where the traverse motor is running forward.)

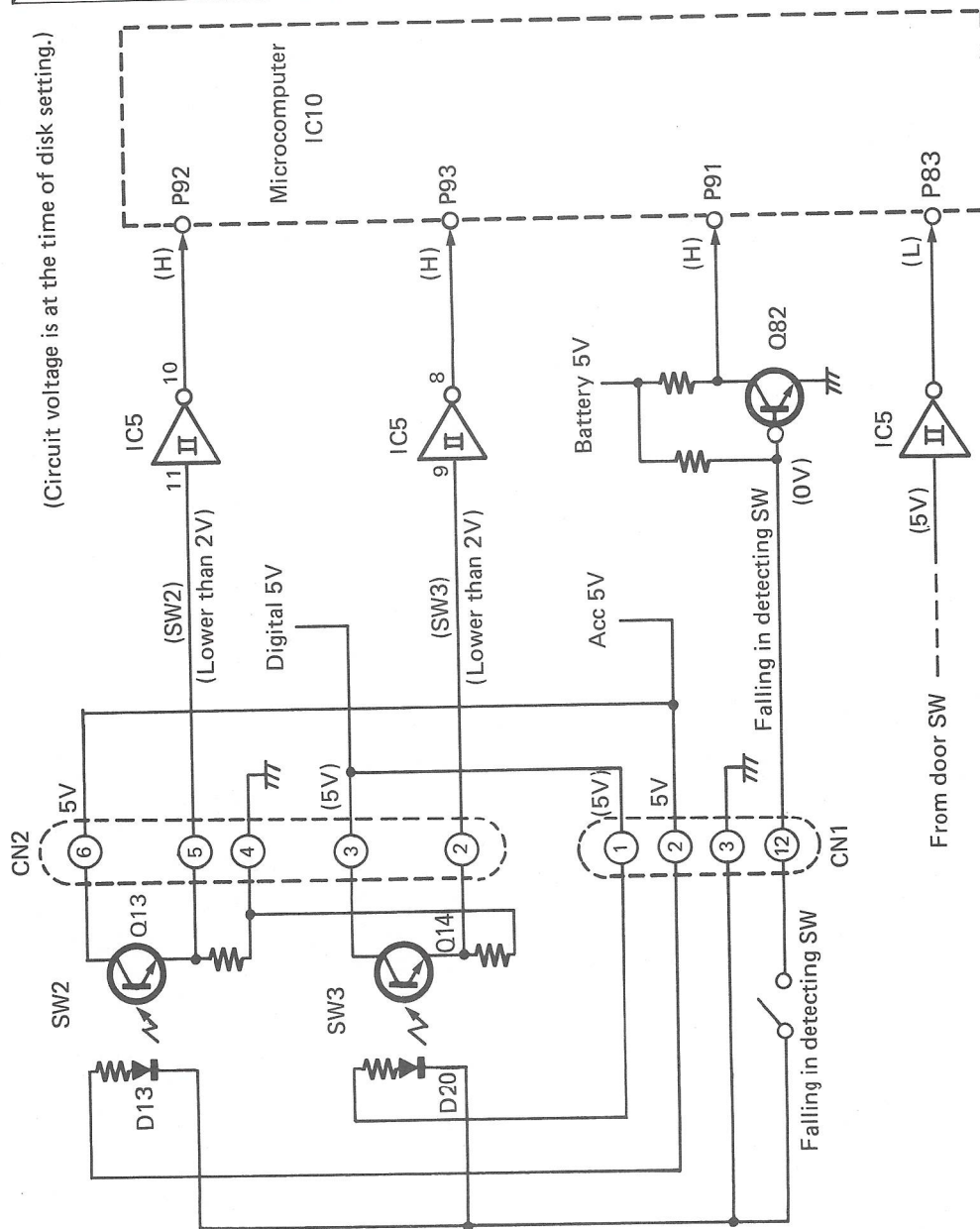
## 2-2 Disk Sensor Check (Sensor Switch)

For a disk to be set on the turntable, the sensor must detect that the disk has been properly inserted so that the disk may be placed on the turntable and held down by the magnet. Check the loading sensor for its detecting operation.

See the Loading and Disk Sensor Circuit

OK

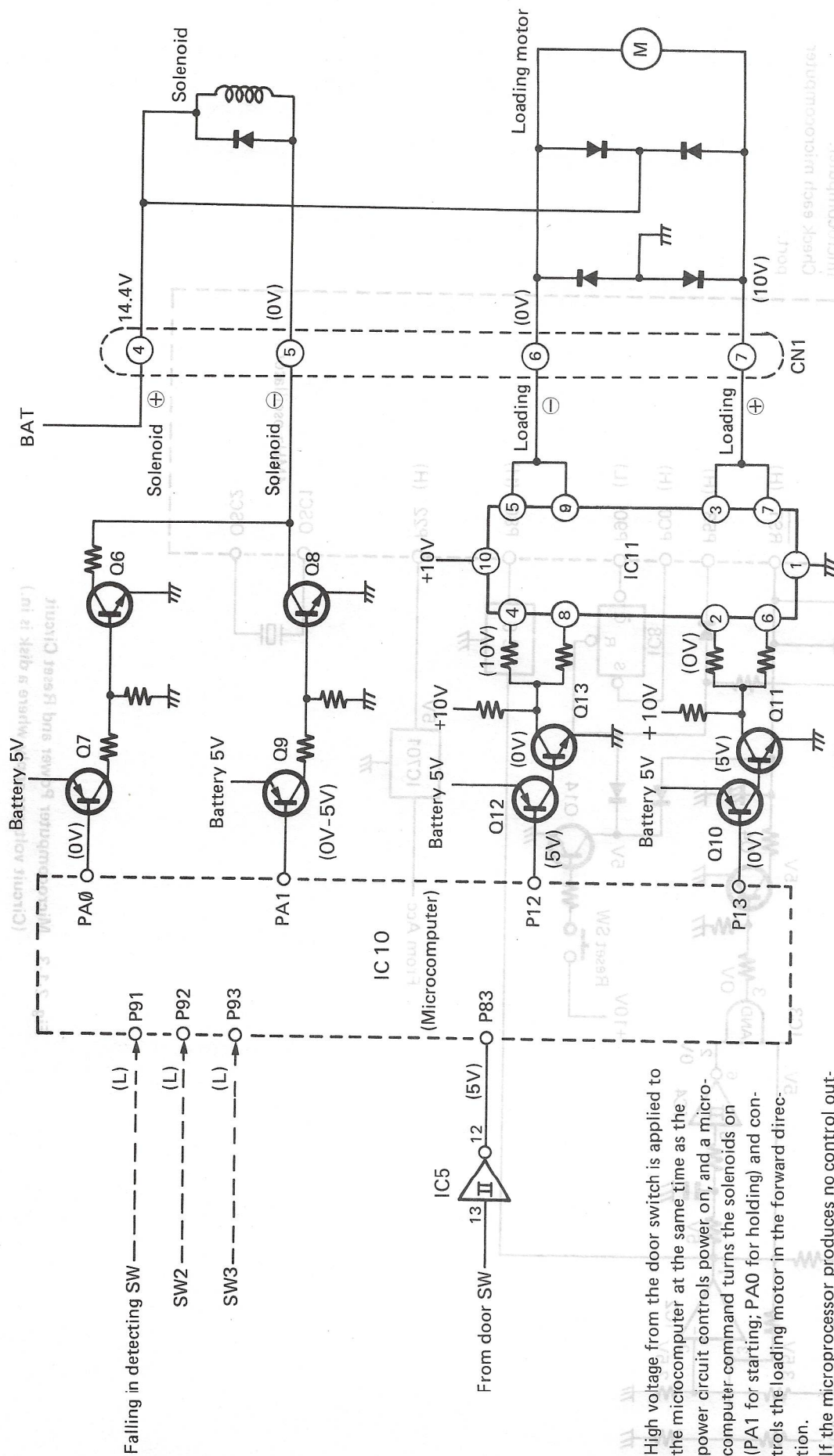
Proceed to 2-3.



\*When a disk is loaded, SW2 turns on the turn SW3 on and turn the solenoid off. Thus, the disk lowers and is held in position on the turntable by the magnet. If detecting circuit is not in good order, it becomes EJECT MODE.

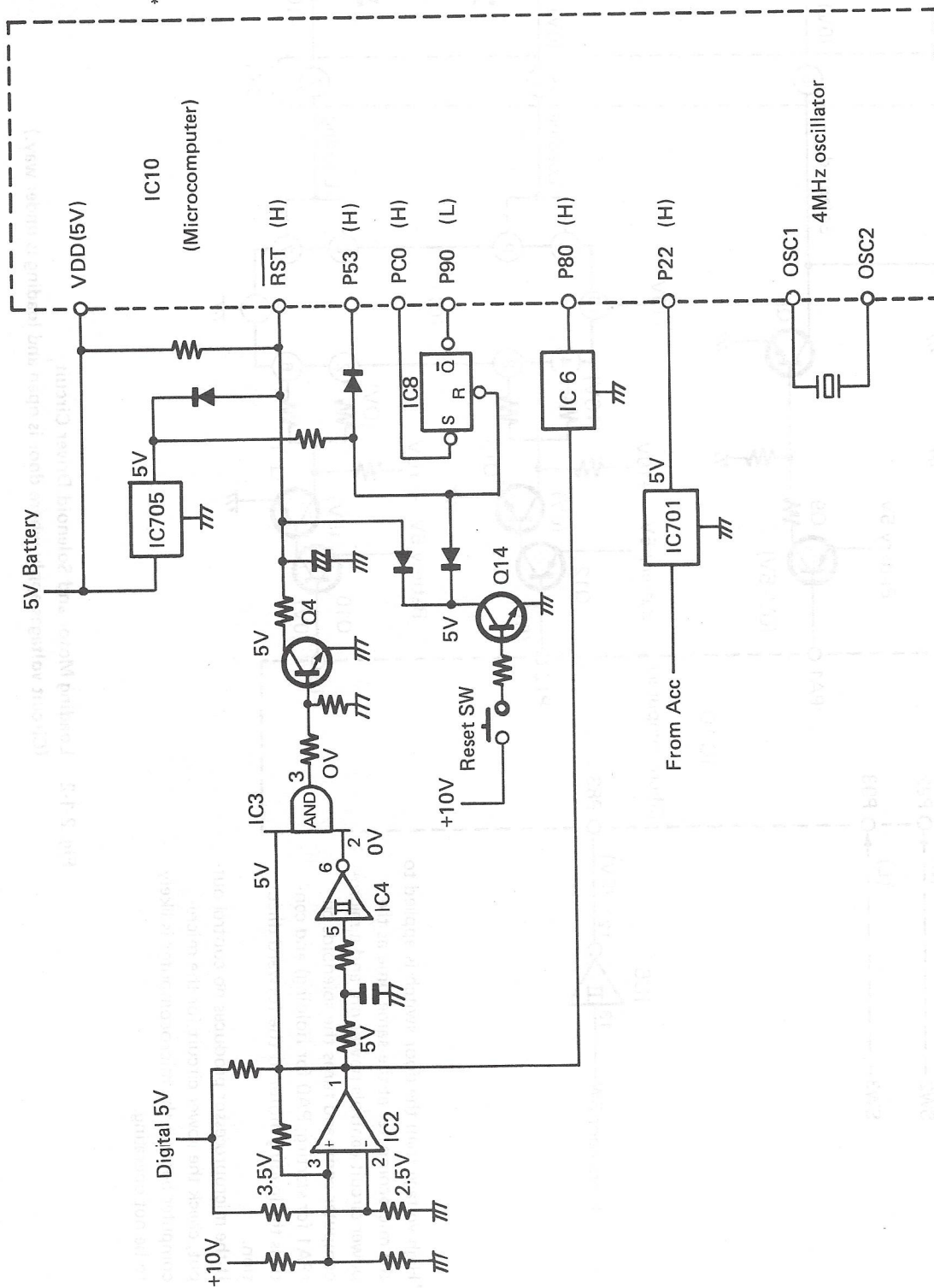
| Mode               | Disk off | Door open | Door closed | Disk set |
|--------------------|----------|-----------|-------------|----------|
| Micro-computer pin |          |           |             |          |
| P92                | L        | L         | H           | H        |
| P93                | H        | L         | H           | H        |
| P91                | L        | L         | L           | H        |
| P83                | L        | H         | L           | L        |

Fig. 2-2-1 Loading and Disk Sensor Circuit (Circuit voltages apply where a disk is set.)



\*High voltage from the door switch is applied to the microcomputer at the same time as the power circuit controls power on, and a microcomputer command turns the solenoids on (PA1 for starting; PA0 for holding) and controls the loading motor in the forward direction. If the microprocessor produces no control output, check the power circuit for the microcomputer because the microcomputer is likely to be not operating.

Fig. 2-1-2 Loading Motor and Solenoid Driver Circuit  
(Circuit voltages apply where door is open and loading is under way.)



\*This circuit diagram shows the power supply for operating microcomputer. Check each microcomputer port.

Fig. 2-1-3 Microcomputer Power and Reset Circuit  
(Circuit voltages apply where a disk is in.)

## 2. CHECKING PROCEDURE

### 2-1 Loading Check (Power unit and microcomputer)

For a disk to be drawn in, the door sensor must detect it to turn power on and the loading motor and solenoid must operate under microcomputer control.

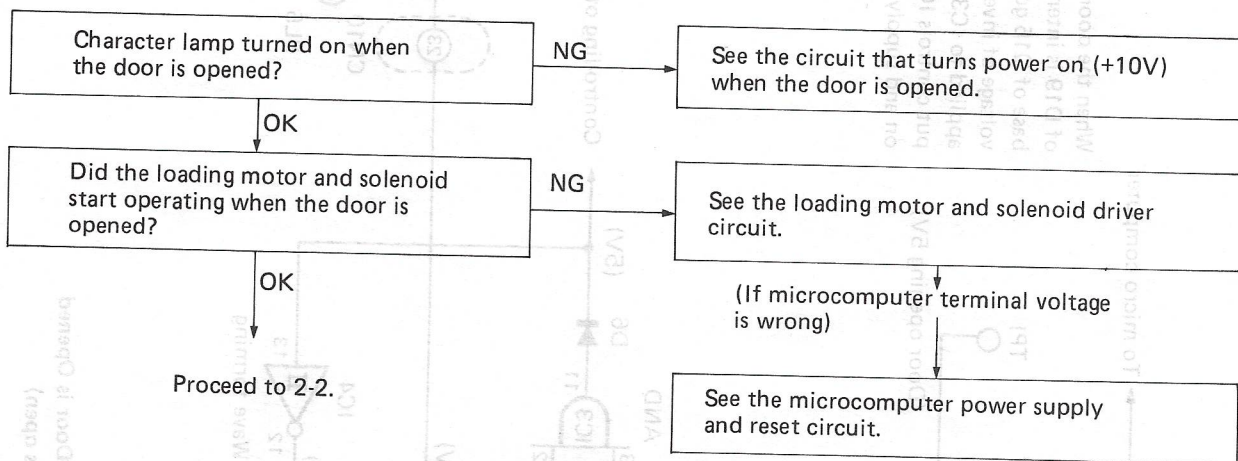
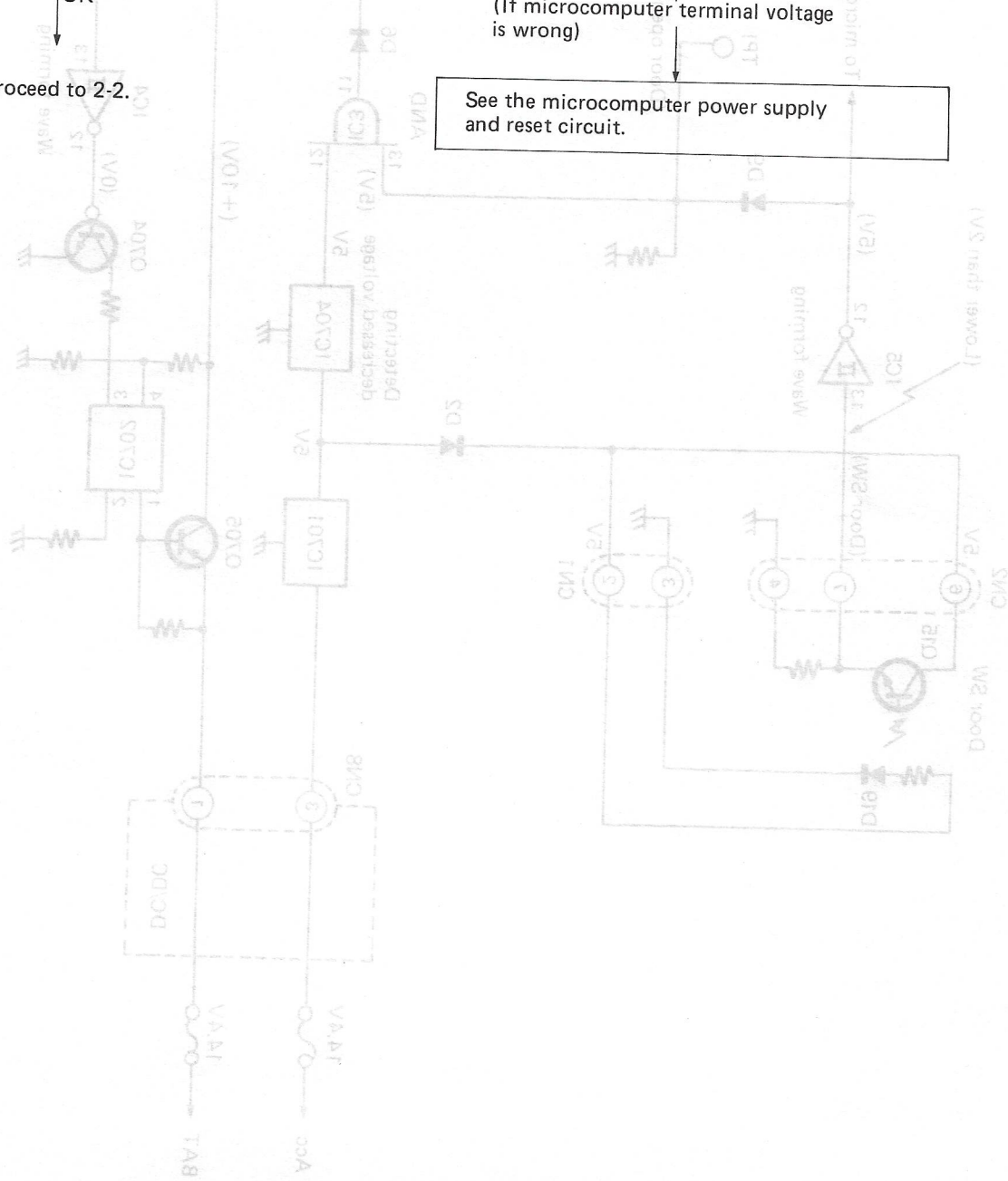


Fig. 5-1-1 Circuit diagram of the door sensor and loading motor/solenoid driver.





If trouble has occurred, check the symptoms according to the flowchart shown in Section 1.  
 Note: Do not skip steps in the flow of symptom confirmation.  
 Keep power on and, if the set is disassembled for troubleshooting, make sure that the symptoms are the same as before disassembling the set.  
 Note: When checking the set with the top cover open, cover the top of the deck to prevent the optical sensor from wrong operation.

## 1. SYMPTOM CHECK

## TROUBLESHOOTING



## TROUBLESHOOTING

If trouble has occurred, check the symptoms according to the flowchart shown in Section 1.

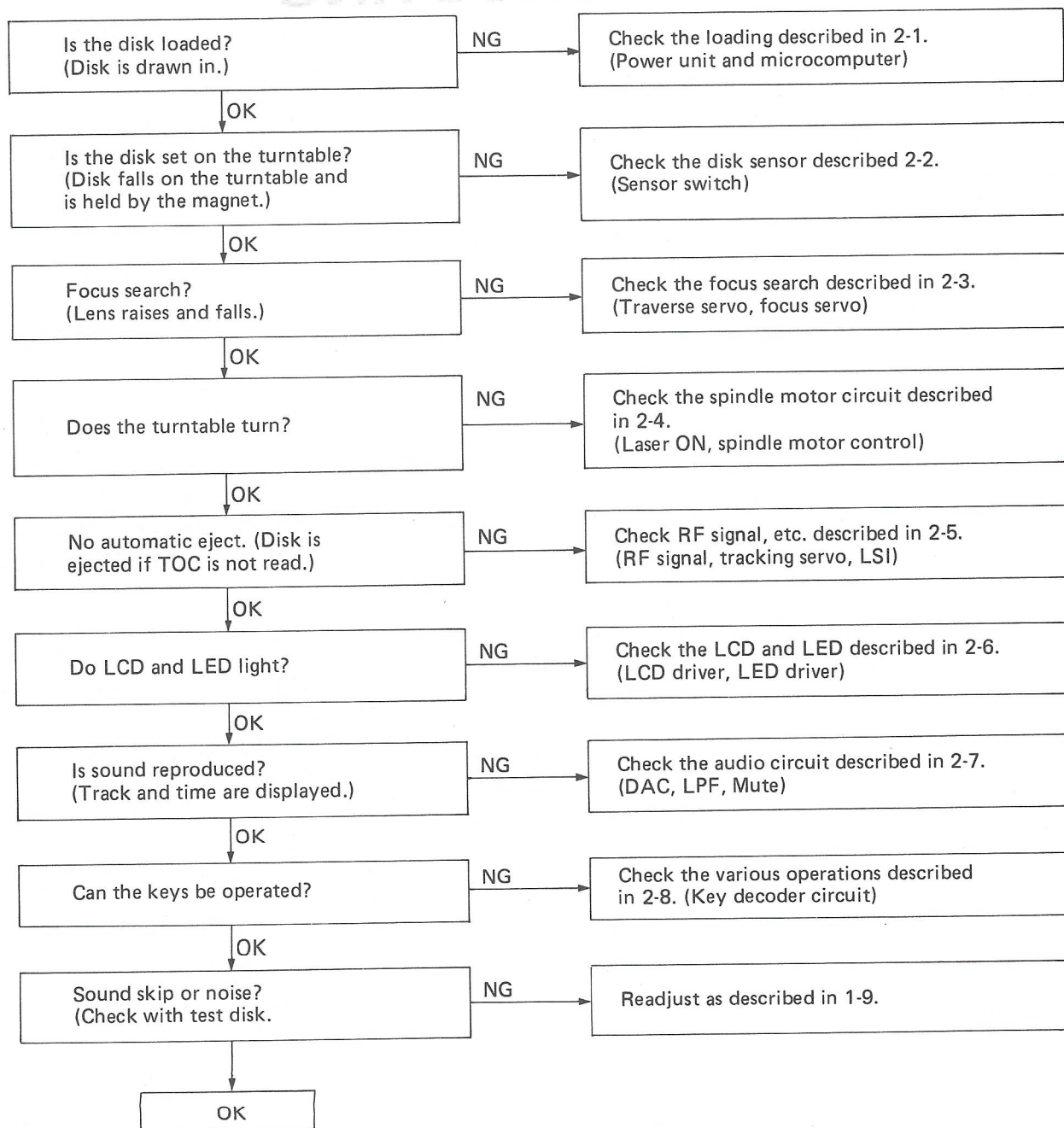
**Note:** Do not skip steps in the flow of symptom confirmation.

Keep power on and, if the set is disassembled for troubleshooting, make sure that the symptoms are the same as before disassembling the set.

**Note:** When checking the set with the top cover open, cover the top of the deck to prevent the optical sensor from wrong operation.

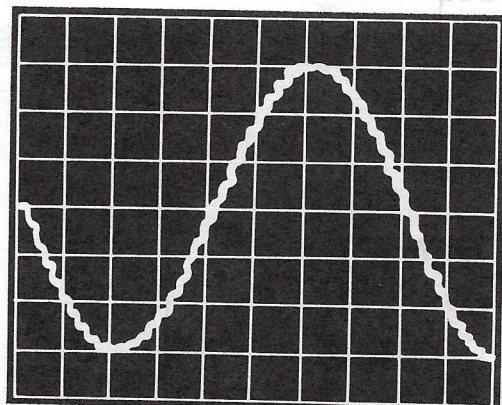
### 1. SYMPTOM CHECK

(Insert a disk.)

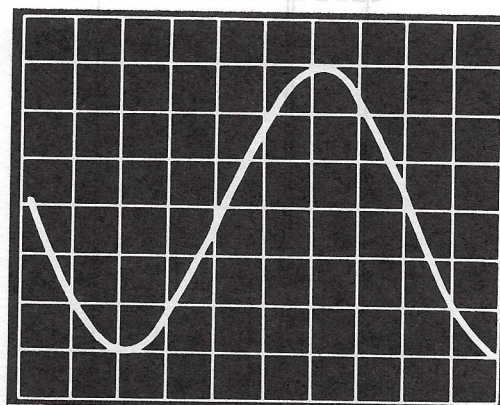


### 10-3 LPF and De-emphasis Circuit

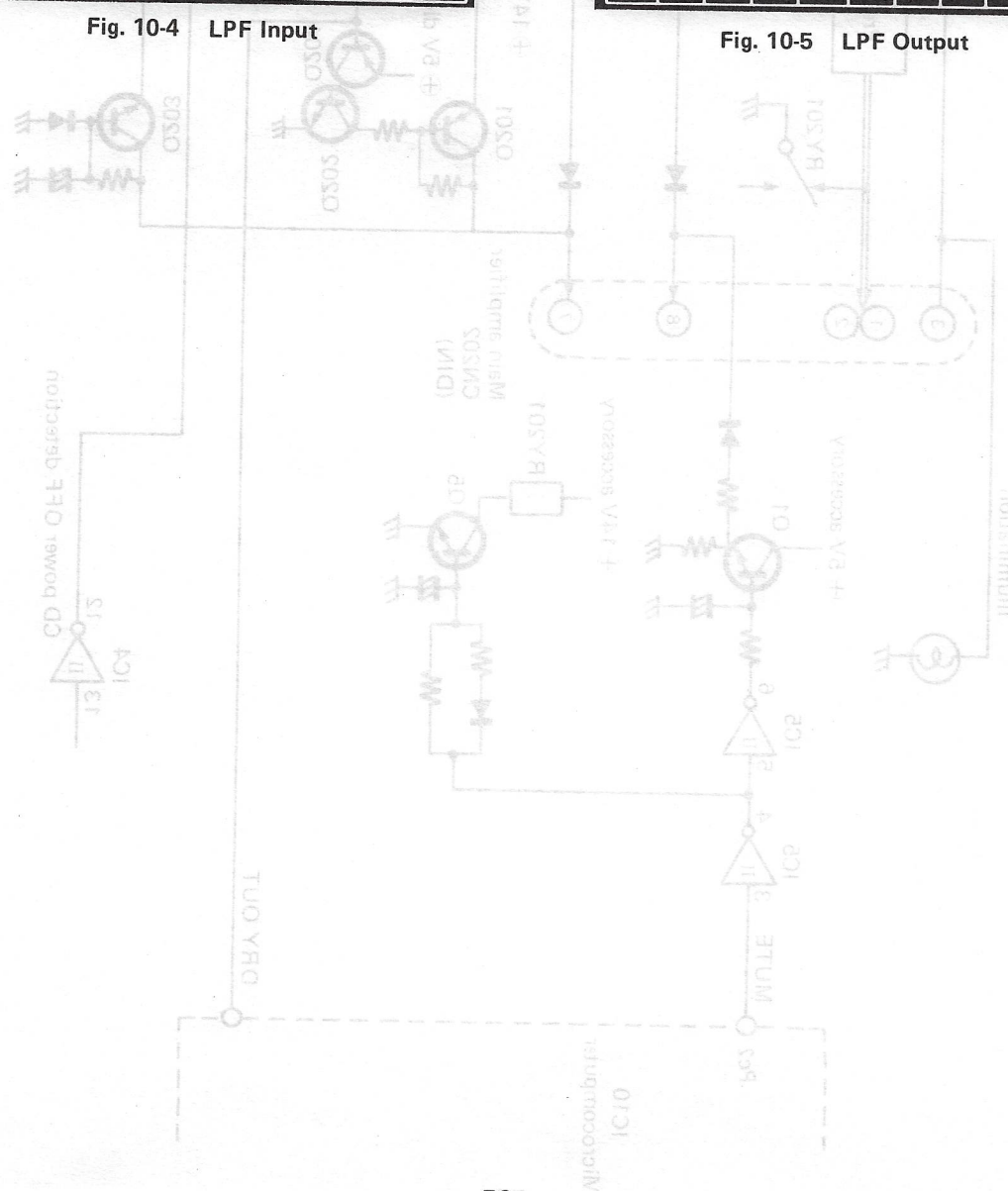
The lowpass filter (LPF) restores the signals containing harmonics (staircase waves) received from the deglitch circuit to complete audio signals. The LPF also has a de-emphasis function. It reads emphasis or no emphasis from subcode (Q channel), and outputs a control signal from pin 21 (DEMPHA) of IC601 (LSI). De-emphasis is applied when the output signal is high.



**Fig. 10-4 LPF Input**



**Fig. 10-5 LPF Output**



# 11. MUTING CIRCUIT and SYSTEM CONTROL

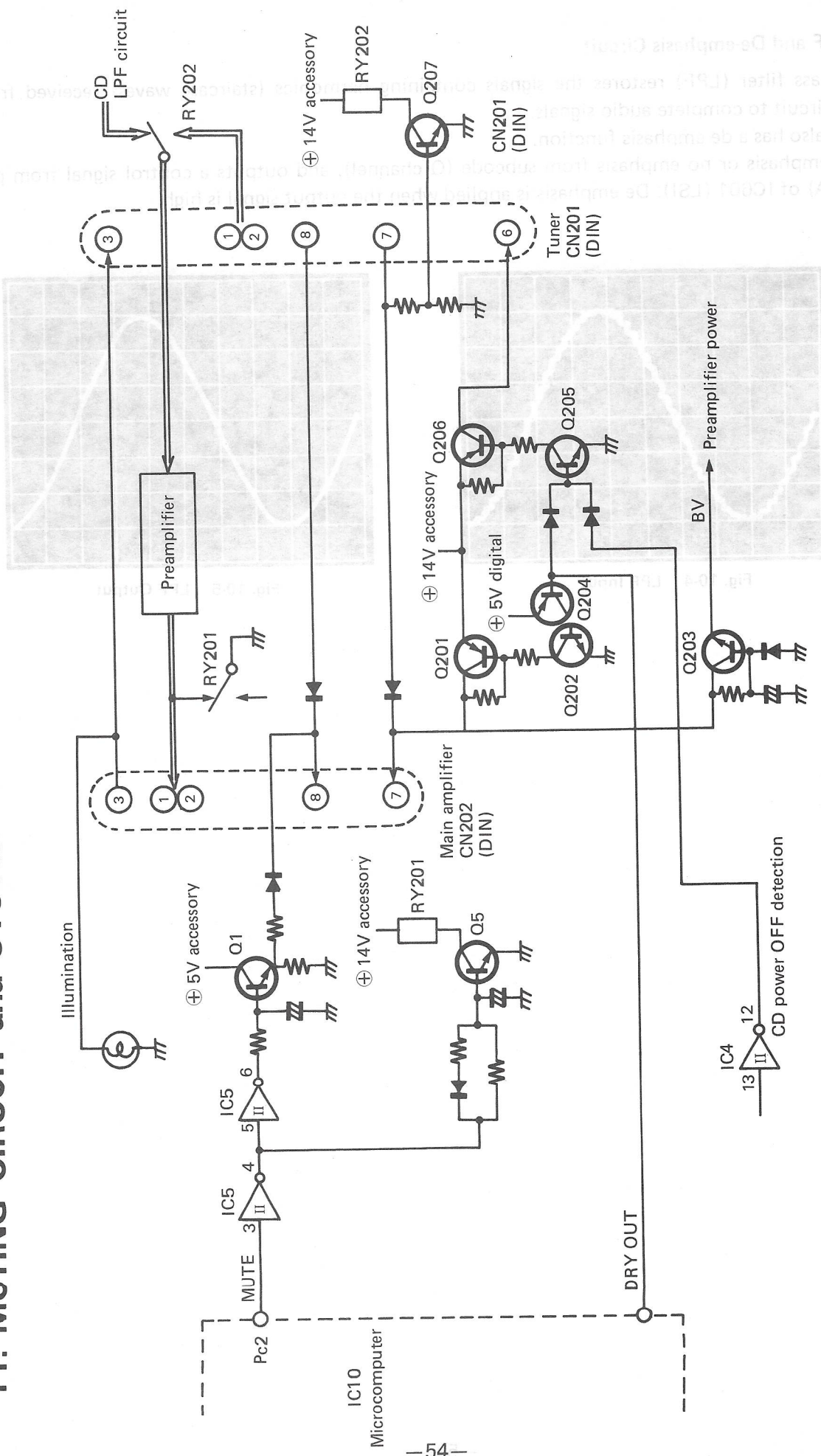


Fig. 11-1 Muting and System Control Circuit

# 10 - AUDIO CIRCUIT

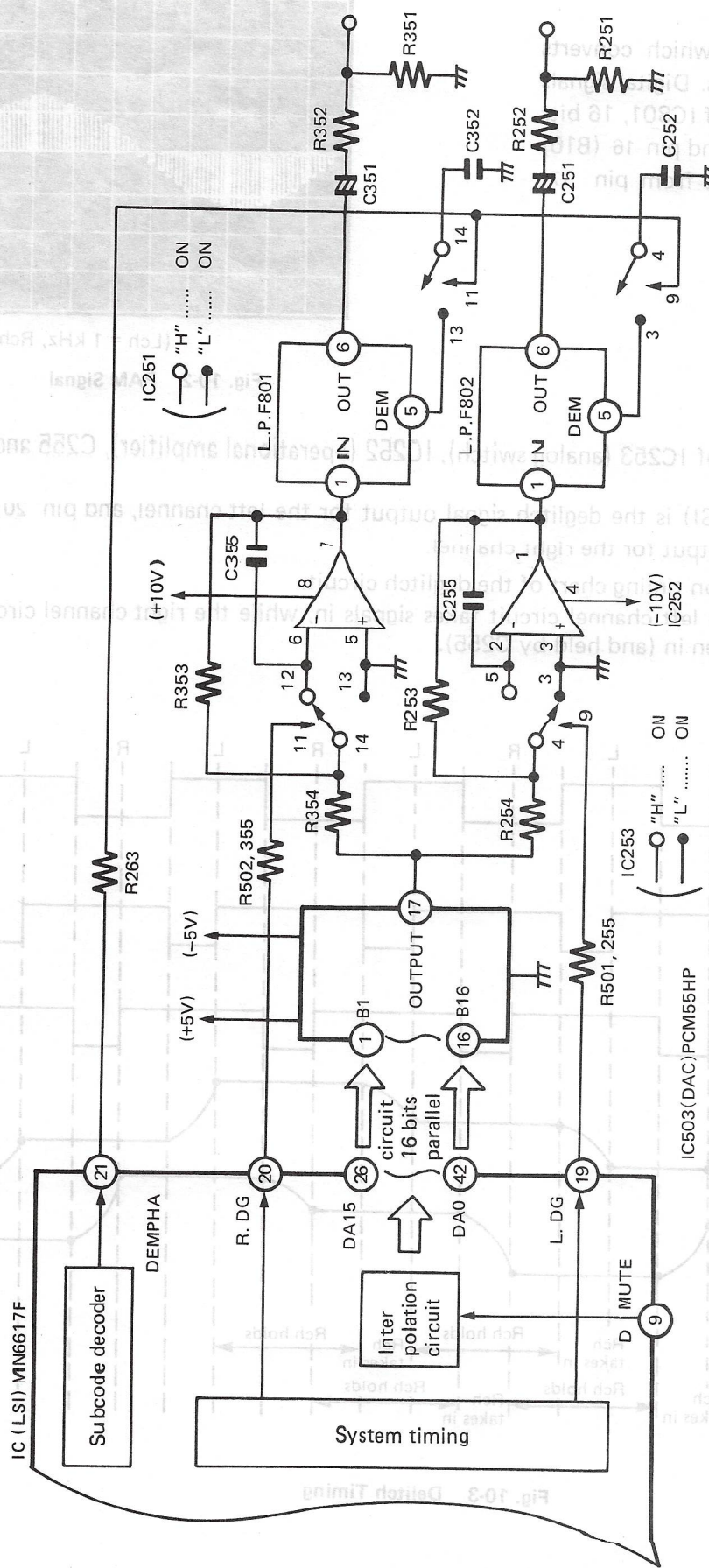
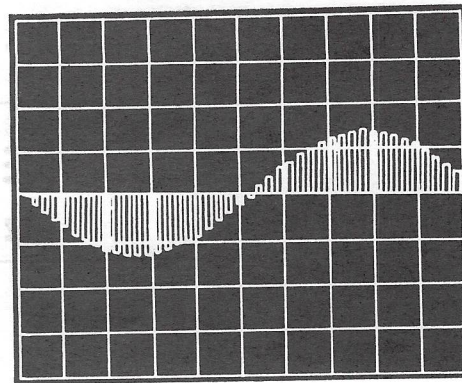


Fig. 10-1 Audio Circuit

## 10-1 D-A Converter

IC801 is a D-A converter, which converts digital signals to analog signals. Digital signals are input to pins 1 to 16 of IC801, 16 bits parallel. Pin 1 (B1) is MSB, and pin 16 (B16) is LSB. PAM signal is output from pin 17 (OUT).



(Lch = 1 kHz, Rch = 0)

Fig. 10-2 PAM Signal

## 10-2 Deglitch Circuit

The deglitch circuit consists of IC253 (analog switch), IC252 (operational amplifier), C255 and C355 (for hold).

Pin 19 (L DG) of IC601 (LSI) is the deglitch signal output for the left channel, and pin 20 (R DG) of IC601 is the deglitch signal output for the right channel.

Fig. 10-3 shows a demodulation timing chart of the deglitch circuit.

When L DG signal is low, the left channel circuit takes signals in, while the right channel circuit outputs the signals that have been taken in (and held by C255).

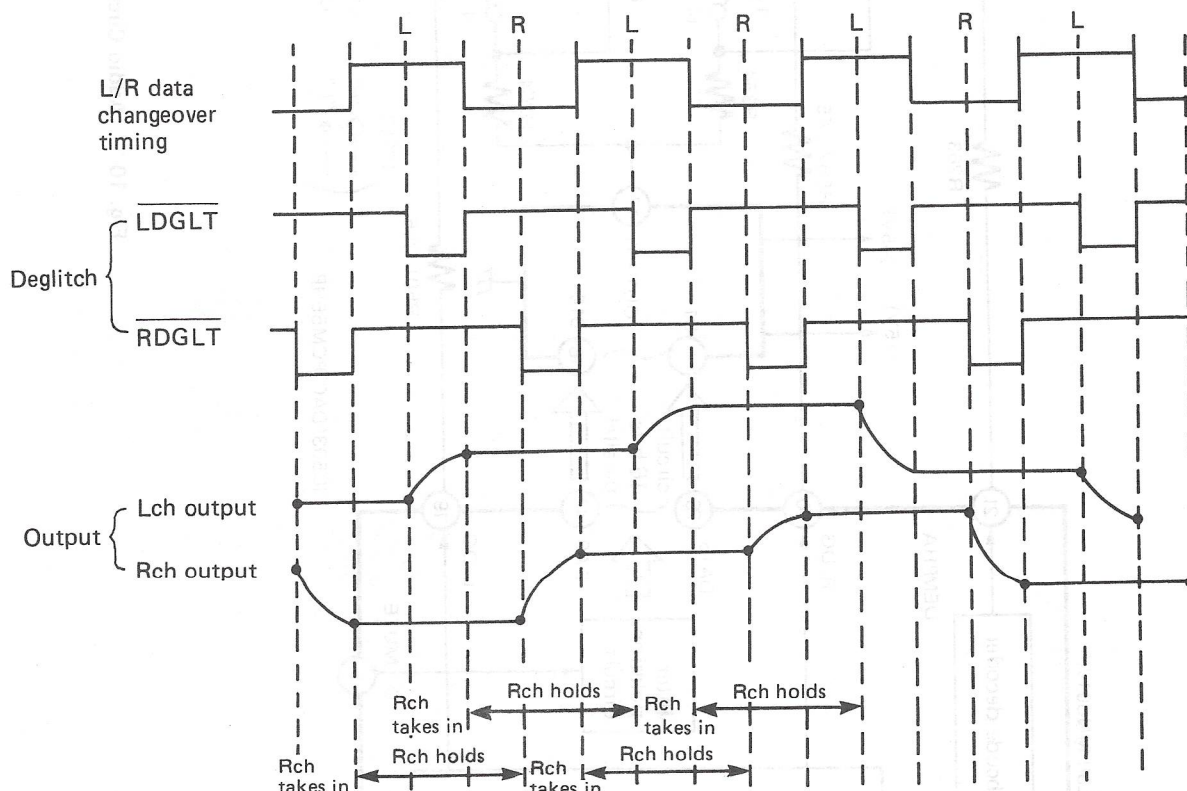


Fig. 10-3 Delitch Timing

### (3) CLV servo

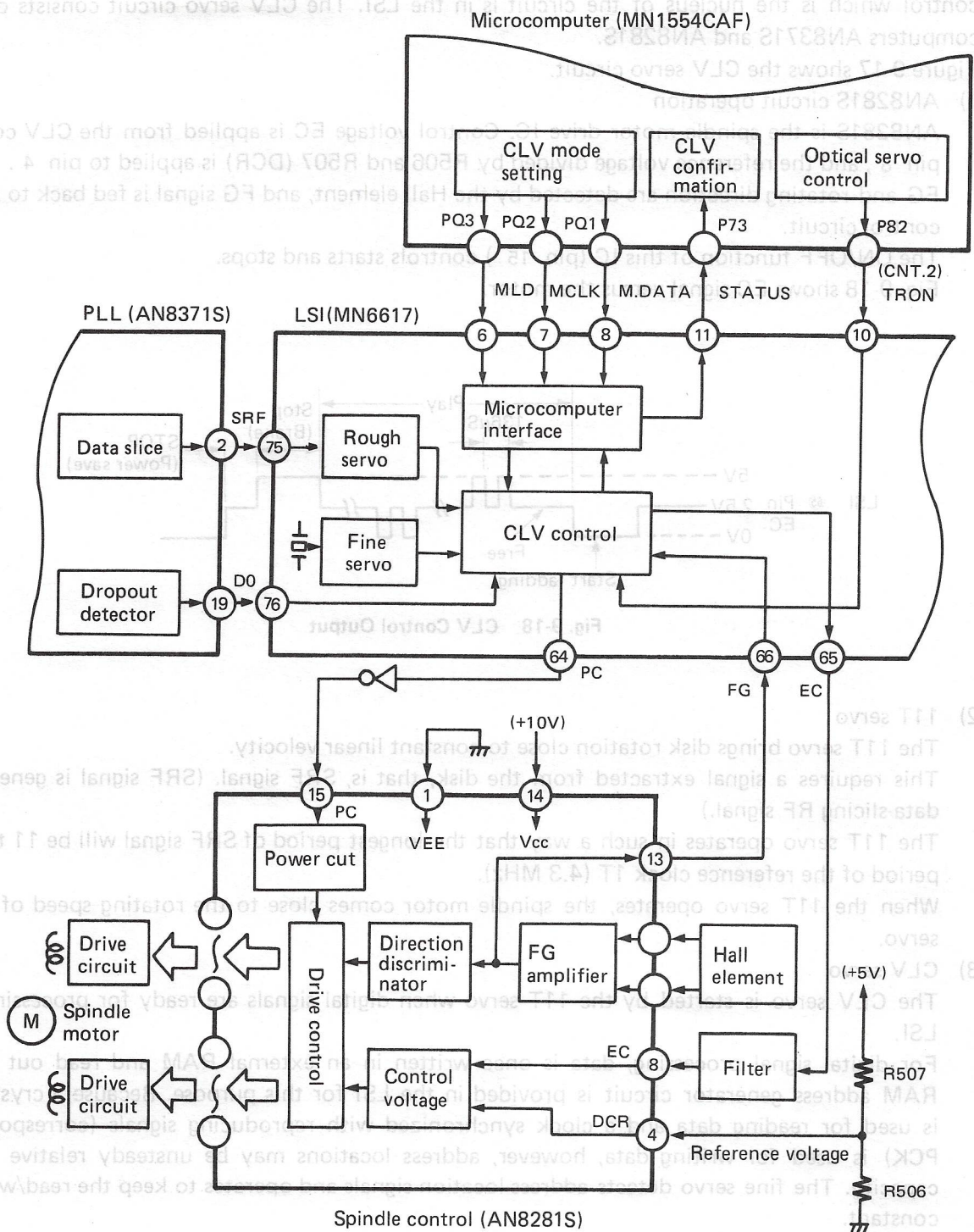


Fig. 9-17 CLV Servo Circuit

The CLV servo is a rotary servo which keeps disk linear velocity constant.

The servo is divided into three parts, that is 11T servo, CLV servo, and traverse tracking servo. The traverse tracking servo and 11T servo are rough servos, and the CLV servo is a fine servo. The CLV control which is the nucleus of the circuit is in the LSI. The CLV servo circuit consists of micro-computers AN8371S and AN8281S.

Figure 9-17 shows the CLV servo circuit.

#### 1) AN8281S circuit operation

AN8281S is the spindle motor drive IC. Control voltage EC is applied from the CLV control to pin 8, and the reference voltage divided by R506 and R507 (DCR) is applied to pin 4.

FG and rotating direction are detected by the Hall element, and FG signal is fed back to the CLV control circuit.

The ON/OFF function of this IC (pin 15) controls starts and stops.

Fig. 9-18 shows EC signal versus the motor.

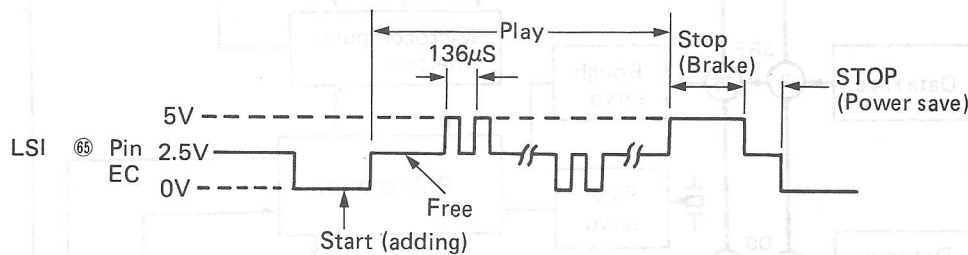


Fig. 9-18 CLV Control Output

#### 2) 11T servo

The 11T servo brings disk rotation close to constant linear velocity.

This requires a signal extracted from the disk, that is, SRF signal. (SRF signal is generated by data-slicing RF signal.)

The 11T servo operates in such a way that the longest period of SRF signal will be 11 times the period of the reference clock 1T (4.3 MHz).

When the 11T servo operates, the spindle motor comes close to the rotating speed of the fine servo.

#### 3) CLV servo

The CLV servo is started by the 11T servo when digital signals are ready for processing in the LSI.

For digital signal processing, data is once written in an external RAM and read out again. A RAM address generator circuit is provided in the LSI for this purpose. Because a crystal clock is used for reading data and a clock synchronized with reproducing signals (corresponding to PCK) is used for writing data, however, address locations may be unsteady relative to RAM capacity. The fine servo detects address location signals and operates to keep the read/write pace constant.

#### 4) Traverse tracking servo

The traverse tracking servo tracks the traverse position to keep linear velocity constant when the traverse mechanism is in rapid feed.

This servo operates in such a way that two consecutive shortest period of SRF will be 6 times the period of the reference clock 1T (4.3 MHz).

TRON is used for switching the traverse tracking servo to the 11T servo or vice versa. That is, the traverse tracking servo is selected when TRON is high; the 11T servo is selected when TRON is low.